

# Senior AI Engineer Interview Guide (Project Terminus / Snorkel AI Based)

This guide is prepared for conducting Senior AI Engineer interviews based on the Project Terminus onboarding slides. It includes: Required skills Technical interview questions Expected answers Practical solutions and scenarios Terminal, Docker, AI task creation, testing, and CI concepts

## Required Skills for Senior AI Engineer

| Skill Area               | Why It Is Important                                                |
|--------------------------|--------------------------------------------------------------------|
| Python & Shell Scripting | Needed for writing solutions, tests, automation, and CLI tasks.    |
| Docker & Containers      | Required for environment setup and reproducible AI task execution. |
| Linux / CLI              | Project tasks are terminal-based and command-driven.               |
| Testing & Pytest         | Needed for deterministic unit tests and validation.                |
| AI Agent Evaluation      | Understanding how LLMs are benchmarked and tested.                 |
| CI/CD Knowledge          | Required for CI checks, linting, validation, and automation.       |
| Debugging                | Important for solving task failures and environment issues.        |
| API Integration          | Useful for tasks involving external services and APIs.             |
| Task Design              | Needed to create multi-step realistic engineering tasks.           |
| Prompt Engineering       | Helps create natural instructions for AI agents.                   |
| Data Processing          | Useful in ML and benchmark-related tasks.                          |

## Senior AI Engineer Interview Questions with Answers

**Question:** 1. What is Terminal-Bench and why is it important?

**Expected Answer:** Terminal-Bench is a benchmark used to evaluate how well AI agents complete complex terminal-based tasks. It helps measure real engineering capabilities of AI systems.

**What You Should Look For:** A strong engineer should explain benchmarking, automation, CLI interaction, and evaluation pipelines.

**Question:** 2. Why are Docker containers used in AI projects?

**Expected Answer:** Docker provides isolated and reproducible environments so tasks behave consistently across systems.

**What You Should Look For:** Candidate should know Dockerfile, docker-compose, dependencies, containers, and environment setup.

**Question:** 3. What makes a good AI evaluation task?

**Expected Answer:** A task should be multi-step, testable, deterministic, realistic, and difficult enough that SOTA models cannot solve it perfectly.

**What You Should Look For:** Look for understanding of benchmarking, edge cases, validations, and task difficulty.

**Question:** 4. Explain the role of Oracle Solution in this project.

**Expected Answer:** Oracle Solution is the expert-written solution script that proves the task is solvable.

**What You Should Look For:** Candidate should know shell scripting and reproducible execution.

**Question:** 5. Why are unit tests important?

**Expected Answer:** Unit tests verify whether the agent completed the task correctly and ensure deterministic validation.

**What You Should Look For:** Expected concepts: pytest, assertions, test coverage, validation logic.

**Question:** 6. What is CI/CD and why is it important?

**Expected Answer:** CI/CD automates testing and validation to maintain code quality and prevent broken submissions.

**What You Should Look For:** Candidate should know linting, pipelines, automated checks, and deployment workflows.

**Question:** 7. How would you prevent cheating in AI evaluation tasks?

**Expected Answer:** Hide tests from the Docker image, avoid hardcoded answers, and use deterministic validation.

**What You Should Look For:** Look for security mindset and benchmark integrity understanding.

**Question:** 8. What are milestones in AI task workflows?

**Expected Answer:** Milestones divide a complex task into smaller sequential stages.

**What You Should Look For:** Candidate should explain modular execution and staged validation.

**Question:** 9. How do you debug failing AI tasks?

**Expected Answer:** Check logs, reproduce failures locally, verify dependencies, inspect tests, and validate environment setup.

**What You Should Look For:** Look for systematic debugging methodology.

**Question:** 10. Why should dependencies be pinned in Dockerfiles?

**Expected Answer:** Pinned dependencies ensure reproducibility and prevent version conflicts.

**What You Should Look For:** Candidate should understand stable builds and environment consistency.

## Practical Scenario-Based Questions

| Scenario                              | Expected Senior-Level Response                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------|
| CI checks are failing.                | Check logs, identify failing validators, reproduce locally, fix task.toml or test cases. |
| Docker container is not starting.     | Inspect Dockerfile, dependencies, ports, permissions, and container logs.                |
| Agent passes incorrectly by cheating. | Hide tests, avoid hardcoded outputs, improve validations.                                |
| Task is too easy for GPT models.      | Increase complexity, add multi-step reasoning, include edge cases.                       |

# Technical Topics Candidate Should Know

Python scripting Shell scripting Docker & docker-compose Linux terminal commands Pytest & unit testing  
Benchmarking AI agents Prompt engineering CI/CD pipelines Debugging techniques Task orchestration API  
integrations Environment setup AI evaluation systems

## **Interview Tip:**

For a Senior AI Engineer role, focus on: Problem-solving approach System design thinking Debugging skills  
Automation mindset Real-world engineering experience Ability to design reliable AI workflows